

Convergence of SMACOF

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TBD

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1 Introduction

The (Euclidean, metric, least squares, p -dimensional) multidimensional scaling (MDS) problem is the minimization of the real valued loss function ¹

$$\sigma(X) := \sum_{1 \leq i < j \leq n} w_{ij} (\delta_{ij} - d_{ij}(X))^2, \quad (1)$$

over *configurations* $X \in \mathcal{X}$, where *configuration space* $\mathcal{X}$ is a subset of $\mathbb{R}^{n \times p}$, the space of all $n \times p$ matrices, with $1 < p < n - 1$. If $\mathcal{X}$ is all of $\mathbb{R}^{n \times p}$ the MDS problem is *unconstrained*, if $\mathcal{X}$ is a proper subset it is *constrained*.

In (1) the *dissimilarities* Δ and *weights* W are the *data* of the problem. Both Δ and W are symmetric, non-negative, and hollow² matrices Δ of order n . The $d_{ij}(X)$ are Euclidean distances between rows x_i and x_j of X . Thus

$$d_{ij}(X) := \sqrt{(x_i - x_j)'(x_i - x_j)}. \quad (2)$$

We call $\sigma(X)$ the *stress* of configuration X (Kruskal (1964a), Kruskal (1964b)). Kruskal does not use weights in his definition and calls our (1) *raw stress*. Since he is only interested in non-metric MDS his definition of stress (or *normalized stress*) adds the sum of squared distances as a denominator in (1).

In this paper we assume that the *dimensionality* p satisfies $1 < p < n - 1$. Both $p = 1$ and $p = n - 1$ are excluded, because these boundary cases of MDS have special properties which makes them fundamentally different from MDS for intermediate values of p (De Leeuw (2023b), De Leeuw et al. (2016)). We also assume, without loss of generality, that W is *irreducible*, which means in this context that the MDS problem does not separate into a number of smaller MDS problems that can be solved independently (De Leeuw (1977)).

¹The symbol $:=$ is used for definitions.

²A hollow matrix has a zero diagonal.

2 Notation

First some convenient notation, introduced in De Leeuw (1977). A *unit vector* e_i has n elements, with element i equal to $+1$, and all other elements zero.

$$A_{ij} := (e_i - e_j)(e_i - e_j)', \quad (3)$$

which means elements (i, i) and (j, j) are equal to $+1$, while (i, j) and (j, i) are -1 . All other elements are zero. With this definition

$$d_{ij}^2(X) = \text{tr } X' A_{ij} X. \quad (4)$$

We also define the matrix

$$V := \sum_{1 \leq i < j \leq n} w_{ij} A_{ij}, \quad (5)$$

and the matrix-valued function B with

$$B(X) := \sum_{1 \leq i < j \leq n} w_{ij} r_{ij}(X) A_{ij}, \quad (6)$$

where

$$r_{ij}(X) := \begin{cases} \frac{\delta_{ij}}{d_{ij}(X)} & \text{if } d_{ij}(X) > 0, \\ 0 & \text{if } d_{ij}(X) = 0. \end{cases} \quad (7)$$

Matrix V is symmetric with non-positive off-diagonal elements. It is doubly-centered (rows and columns add up to zero) and thus weakly diagonally dominant. It follows that it is positive semi-definite (see Varga (1962), section 1.5). Because of irreducibility it has rank $n - 1$, and the vectors in its null space are all proportional to e , the vector with all elements equal to $+1$. For all X the matrix $B(X)$ is also symmetric, positive semi-definite, and doubly-centered.

If we assume, again without any real loss of generality, that

$$\sum_{1 \leq i < j \leq n} w_{ij} \delta_{ij}^2 = 1, \quad (8)$$

then

$$\sigma(X) = 1 - 2\text{tr } X' B(X) X + \text{tr } X' V X. \quad (9)$$

Still following De Leeuw (1977) we also define

$$\rho(X) := \sum_{1 \leq i < j \leq n} w_{ij} \delta_{ij} d_{ij}(X) = \text{tr } X' B(X) X, \quad (10)$$

$$\eta^2(X) := \sum_{1 \leq i < j \leq n} w_{ij} d_{ij}^2(X) = \text{tr } X' V X, \quad (11)$$

and

$$\lambda(X) := \frac{\rho(X)}{\eta(X)}. \quad (12)$$

Thus

$$\sigma(X) = 1 - 2\rho(X) + \eta^2(X), \quad (13)$$

and

$$\min_{\alpha} \sigma(\alpha X) = 1 - \lambda^2(X). \quad (14)$$

3 SMACOF

The Cauchy-Schwartz inequality implies that for any pair (X, Y) of configurations

$$d_{ij}(X) \geq \frac{1}{d_{ij}(Y)} \text{tr } X' A_{ij} Y, \quad (15)$$

from which

$$\rho(X) \geq \text{tr } X' B(Y) Y. \quad (16)$$

with equality if $X = Y$.

Define the *Guttman Transform* of configuration X as

$$\Gamma(X) := V^+ B(X) X, \quad (17)$$

with V^+ the Moore-Penrose inverse of V given by

$$V^+ = (V + \frac{1}{n} ee')^{-1} - \frac{1}{n} ee'. \quad (18)$$

Then, from (13) and (16),

$$\begin{aligned} \sigma(X) &\leq 1 - 2\text{tr } X' B(Y) Y + \text{tr } X' V X = \\ &= 1 + \text{tr } (X - \Gamma(Y))' V (X - \Gamma(Y)) - \text{tr } \Gamma(Y)' V \Gamma(Y), \end{aligned} \quad (19)$$

with equality if $X = Y$. It follows that if we choose X so that

$$\text{tr } (X - \Gamma(Y))' V (X - \Gamma(Y)) \leq \text{tr } (Y - \Gamma(Y))' V (Y - \Gamma(Y)), \quad (20)$$

then $\sigma(X) \leq \sigma(Y)$. If there is strict inequality in (20) then $\sigma(X) < \sigma(Y)$. In particular if $Y \neq \Gamma(Y)$ then $\sigma(\Gamma(Y)) < 1 - \eta^2(\Gamma(Y)) = \sigma(Y)$.

The *SMACOF algorithm* for unconstrained MDS is, with a superscripted iteration counter $k \geq 1$,

$$X^{(k)} = \Gamma(X^{(k-1)}) = \Gamma^k(X^0), \quad (21)$$

where $\Gamma^0 = I$ and $\Gamma^k(X) := \Gamma(\Gamma^{k-1}(X))$. If during the iterations we encounter an X for which $X = \Gamma(X)$ we have finite convergence and we stop. If $p > 1$ we can safely assume that the algorithm generates an infinite sequence (i.e. does not stop).

De Leeuw (1977) shows that the SMACOF iterations (39) converge (in a specific sense described later in this paper) to a fixed point of the Guttman transform, i.e. to an X with $\Gamma(X) = X$. If stress is differentiable at X then the fixed points of the Guttman transform are stationary points, where the derivative of stress vanishes. De Leeuw (1984) shows that at a

local minimum we have $d_{ij}(X) > 0$ for all (i, j) with $w_{ij}\delta_{ij} > 0$, i.e. stress is differentiable at local minima. The equation $\Gamma(X) = X$ says that the columns of X are eigenvectors of $V^+B(X)$, with eigenvalues equal to one. De Leeuw (2014) shows that if $\Gamma(X)X = X$ and all eigenvalues of $V^+B(X)$ are less than or equal to one then stress has its global minimum at X .

The SMACOF algorithm (without weights) was proposed by Guttman (1968) by differentiating stress and setting the derivatives equal to zero. This derivation ignored the problem that stress is not differentiable if one or more distances are zero, and it also did not say anything about convergence of the algorithm. In De Leeuw (1977) and De Leeuw and Heiser (1977) a new derivation of the algorithm was given, using the majorization argument based on the Cauchy-Schwarz inequality. This made it possible to avoid problems with differentiability, and it lead to a simple convergence proof.

4 Global Convergence of SMACOF

The main result in De Leeuw (1977), which relies heavily on theory developed in Zangwill (1969), Ostrowski (1973), and Meyer (1976), is:

1. The iterates $X^{(k)}$ are in the compact set $\eta(X) \leq 1$,
2. $\{\rho(X^k)\}$, $\{\eta(X^k)\}$ and $\{\lambda(X^k)\}$ are increasing sequences, with limits ρ_∞, η_∞ and λ_∞ , where $\eta_\infty := \lambda_\infty := \sqrt{\rho_\infty}$.
3. $\{\sigma(X^k)\}$ is a decreasing sequence converging to $\sigma_\infty := 1 - \eta_\infty^2$.
4. At any accumulation point X_∞ of $\{\sigma(X^k)\}$ we have $\sigma(X_\infty) = \sigma_\infty$ and $X_\infty = \Gamma(X_\infty)$.
5. $\eta(X^{(k+1)} - X^{(k)})$ converges to zero and thus either $\{\sigma(X^k)\}$ converges or the set of accumulation points of $\{\sigma(X^k)\}$ is a continuum³.

So, in words, the sequence of stress values converges monotonically. The sequence of configurations is asymptotically regular⁴ and has one or more accumulation points. Each accumulation point is a fixed point of the Guttman transform. All accumulation points have the same function value, which is also the limit of the stress sequence.

But it is important to emphasize that De Leeuw (1977) did not show that the sequence of configurations $\{X^{(k)}\}$ is actually a Cauchy sequence, and consequently converges to a configuration X_∞ . This is basically because of the rotational indeterminacy of the MDS problem. If K is a rotation matrix, i.e. $K'K = KK' = I$, then $d_{ij}(XK) = d_{ij}(X)$ for all i and j , and thus $\sigma(XK) = \sigma(X)$. If X is a fixed point of the Guttman transform, then $\Gamma(XK) = \Gamma(X)K = XK$, and thus XK is a fixed point as well. The rotations of X define a continuum. There are no isolated fixed points, each fixed point is part of a nonlinear continuum of fixed points. Local minima are never strict.

De Leeuw (1993) shows that stress has only a single local maximum at $X = 0$, and consequently the only possible stationary points are local minima and saddle points. It is shown by De Leeuw (2019) that all fixed points of the Guttman transform which are not of full column rank are saddle points. Thus if X is a local minimizer of MDS in p dimensions, then adding q zero columns to X produces a saddle point $[X \mid 0]$ for MDS in $p + q$ dimensions. At saddle points there are always directions in which stress can be decreased, and consequently the SMACOF algorithm with enough iterations will eventually get close to a continuum of local minima (Lee et al. (2019)).

³A continuum is a closed set which is not a union of two or more disjoint closed sets.

⁴A sequence $\{x_i\}$ is asymptotically regular if $\|x_{i+1} - x_i\| \rightarrow 0$ as $i \rightarrow \infty$.

5 Local Convergence of SMACOF

De Leeuw (1988) studies the local convergence of SMACOF, i.e. the *rate of convergence* of the SMACOF iterations. There is, however, a problem not adequately addressed in that article, which has quite a bit of hand-waiving, and some incorrect statements. If there is no convergence to a single configuration then the usual rate of convergence is not defined either.

We know the sequence $\{\eta(X^{(k+1)} - X^{(k)})\}$ tends to zero. We can measure its rate of convergence by the *ratio factor*

$$q_k = \frac{\eta(X^{(k+1)} - X^{(k)})}{\eta(X^{(k)} - X^{(k-1)})}, \quad (22)$$

and the *root factor*

$$r_k = \eta(X^{(k+1)} - X^{(k)})^{1/k}. \quad (23)$$

There is no guarantee that either $\{q_k\}$ or $\{r_k\}$ converges, but

$$\liminf_{k \rightarrow \infty} q_k \leq \liminf_{k \rightarrow \infty} r_k \leq \limsup_{k \rightarrow \infty} r_k \leq \limsup_{k \rightarrow \infty} q_k \quad (24)$$

Thus if $\lim_{k \rightarrow \infty} q_k$ exists then $\lim_{k \rightarrow \infty} r_k$ exist and the two limit values are equal. See Rudin (1976), p. 68, theorem 3.37.

The rate of convergence of the SMACOF iterations at a fixed point X is computed in De Leeuw (1988) as the spectral norm $\kappa(X) = \|\mathcal{D}\Gamma(X)\|_\infty$, i.e. as the modulus of the largest eigenvalue of the derivative of the Guttman transform. There are good reasons to compute this quantity. From Ostrovski's Theorem (, theorem 22.1, p. 151, Ortega and Rheinboldt (1970), theorem 10.1.3, p. 300) we know that if $\kappa(X) < 1$, then X is a *point of attraction* of the SMACOF iteration. If we start close enough then the iterations will converge to X . We also know (Ostrowski (1973), theorem 22.2, p. 152) that if $\kappa(X) > 1$ then X is a *point of repulsion*, which means the iterations will diverge when started in a certain solid angle with X at its apex.

If we define, with Ortega and Rheinboldt (1970), chapter 9, for a sequence $\{X^{(k)}\}$ converging to X , the ratio and root convergence factors

$$Q_1(\{X^{(k)}\}) = \limsup_{k \rightarrow \infty} \frac{\eta(X^{(k+1)} - X)}{\eta(X^{(k)} - X)}, \quad (25)$$

and

$$R_1(\{X^{(k)}\}) = \limsup_{k \rightarrow \infty} \eta(X^{(k)} - X)^{1/k}, \quad (26)$$

then $\kappa(X) < 1$ implies $R_1(X^{(k)}) = \kappa(X)$ (Ortega and Rheinboldt (1970), theorem 10.1.4, p. 301). Also $R_1(X^{(k)})$ is independent of the norm we have chosen, which is η in the SMACOF

case (Ortega and Rheinboldt (1970), theorem 9.9.2, p. 288). Because $Q_1(X^{(k)})$ depends on the norm we can only assert that $Q_1(X^{(k)}) \geq R_1(X^{(k)}) = \kappa(X)$, although it is guaranteed in the SMACOF case that some norm exists for which there is equality (Ortega and Rheinboldt (1970), Notes and Remarks NR 10.1-5, p. 306).

There is a problem, however, with applying the Ostrowski and Ortega-Rheinboldt results in our case. Because of the invariance of distances under rotation we have $\kappa(X) = 1$, no matter what X is. And, basically for the same reason, we have not shown that $X^{\wedge\{k\}}$ converges to a fixed point at all. De Leeuw (1988) on page 171 acknowledges the problem. In fact, he proposes a way around the problem in the proof of his theorem 3 on page 175, but the proof is not very convincing, although convincing enough to get past the reviewers. I will try to be more specific and precise.

Observe that

$$\mathcal{D}\Gamma(X) = V^+ \mathcal{D}^2 \rho(X), (\#eq : dgamma) \quad (27)$$

and

$$\mathcal{D}^2 \sigma(X) = V - \mathcal{D}^2 \rho(X) = V(I - \mathcal{D}\Gamma(X)). (\#eq : d2sigma) \quad (28)$$

Because ρ is convex $\mathcal{D}^2 \rho(X)$ is symmetric and positive semi-definite. The eigenvalues of $\mathcal{D}\Gamma(X)$ are the generalized eigenvalues of the positive semi-definite matrix pair $(\mathcal{D}^2 \rho(X), V)$.

An explicit formula for the derivatives of the Guttman transform, or equivalently for the second derivatives of stress, was first given by De Leeuw (1988). We write $\partial\Gamma(X; Y)$ for the directional derivative at X in the direction Y . Partial derivatives can be obtained by choosing $Y = e_i e'_s$. Of course we assume here that $d_{ij}(X) > 0$ for all $i < j$ with $w_{ij} \delta_{ij} > 0$.

$$\partial\Gamma(X; Y) = V^+ \{B(X)Y - H(X, Y)X\}, (\#eq : dgamma mat) \quad (29)$$

with

$$H(X, Y) = \sum_{1 \leq i < j \leq n} w_{ij} \frac{\delta_{ij}}{d_{ij}(X)} \frac{\text{tr } X' A_{ij} Y}{d_{ij}^2(X)} A_{ij}. (\#eq : hdef) \quad (30)$$

These formulas, together with equations @ref(eq:dgamma) and @ref(eq:d2sigma), prove the main results in De Leeuw (1988).

1. All eigenvalues of $\mathcal{D}\Gamma(X)$ are non-negative.
2. **(Homogeneity)** X is an eigenvector of $\mathcal{D}\Gamma(X)$ with eigenvalue zero.
3. **(Translation)** $\mathcal{D}\Gamma(X)$ has at least p additional eigenvalues equal to zero, corresponding with eigenvectors of the form ee'_s , where e has all elements equal to one and e_s has one of its elements equal to one and the others equal to zero.
4. **(Rotation)** If X is a fixed point of the Guttman transform then $\mathcal{D}\Gamma(X)$ has at least $\frac{1}{2}p(p-1)$ eigenvalues equal to one, corresponding with eigenvectors of the form XA , with A anti-symmetric.

5. At a local minimum point X of stress all eigenvalues of $\mathcal{D}\Gamma(X)$ are less than or equal to one. At a strict local minimum they are strictly less than one.
6. At a saddle point X of stress the largest eigenvalue of $\mathcal{D}\Gamma(X)$ is larger than one.

6 Rotated SMACOF

6.1 Rotation to Principal Components

To avoid the problems caused by a continuum of fixed points De Leeuw (1988) proposes to rotate each SMACOF iterate to a unique position, for example to principal components. For any X with singular value decomposition $X = K\Lambda L'$ we define $\Pi(X) = K\Lambda = XL$. Note that if one or more singular values are equal then Π is not unique and becomes a point-to-set map. We can rotate within each of the spaces corresponding to multiple singular values, and thus we have not completely eliminated indeterminacy due to rotation. In the sequel we simply assume that the p singular values of X are different.

Now consider the iterations

$$\tilde{X}^{(k+1)} = \Pi(\Gamma(\tilde{X}^{(k)})) = (\Pi\Gamma)^k(\tilde{X}^0). \quad (31)$$

Let's call this rotated SMACOF, or rSMACOF for short.

It looks initially as if rSMACOF requires a great deal of extra computation, but this is actually not the case. If L is a rotation matrix we have $\Pi(XL) = \Pi(X)$ and $\Gamma(XL) = \Gamma(X)L$. This implies that $\Pi\Gamma\Pi(X) = \Pi\Gamma(X)$ for all X . As a result the sequences $\{X^{(k)}\}$ and $\{\tilde{X}^{(k)}\}$ have a simple relationship. If we start the $\{\tilde{X}^{(k)}\}$ sequence with X^0 then for each k we have $\sigma(\tilde{X}^{(k)}) = \sigma(X^{(k)})$ and $\tilde{X}^{(k)} = \Pi(X^{(k)})$. Thus the two sequences of stress values are exactly the same for SMACOF and mSMACOF and the two configurations, and consequently also the accumulation points of the two sequences of configurations, just differ by a rotation. To get $\tilde{X}^{(k)}$ we can just compute the SMACOF iterate $X^{(k)}$ and rotate it to principal components.

De Leeuw (2008)

The derivative of $\Pi\Gamma$ is, by the chain rule,

$$\mathcal{D}\Pi\Gamma(X) = \mathcal{D}\Pi(\Gamma(X))\mathcal{D}\Gamma(X). \quad (32)$$

After some computation we find

$$\mathcal{D}\Pi(X)[Y] = YL + K\Lambda M, \quad (33)$$

where $X = K\Lambda L'$ and M is the anti-symmetric matrix with off-diagonal elements

$$m_{ij} = -\frac{\lambda_i u_{ij} + \lambda_j u_{ji}}{\lambda_i^2 - \lambda_j^2} \quad (34)$$

with $U = K'YL$.

If X satisfies $\Pi(X) = X$ then $L = I$ and thus $\mathcal{D}\Pi(X)[Y] = Y + XM$ and $U = K'Y$. We proceed to compute the eigenvectors and eigenvalues of the Jacobian at an X with $\Pi(X) = X$.

1. If $Y = XA$ with A antisymmetric then $M = -L'AL$ and $\mathcal{D}\Pi(X)[Y] = 0$. This corresponds with $\frac{1}{2}p(p-1)$ zero eigenvalues
2. If $Y = K\Lambda^{-1}A$ with A antisymmetric then $M = 0$. This gives $\frac{1}{2}p(p-1)$ eigenvectors with eigenvalue one.
3. If $Y = K\Lambda^{-1}D$ with D diagonal then $M = 0$. This gives another p eigenvectors with eigenvalue one.
4. If $Y = K_{\perp}S$ with $K_{\perp}$ a basis for the orthogonal complement of X then $M = 0$ and we have another $p(n-p)$ eigenvalues equal to one.

This is a complete set of eigenvectors. It turns out all eigenvalues are equal to one, except for $\frac{1}{2}p(p-1)$ zero eigenvalues. It follows that fixed points of $\Pi\Gamma$ are of the form $\tilde{X} = XL$, with X a fixed point of Γ and L the rotation of X to principal components). Thus $\Gamma(\tilde{X}) = \Gamma(X)L = XL = \tilde{X}$, and $\tilde{X}$ is a fixed point of both Γ and $\Pi\Gamma$. The eigenvalues of $\mathcal{D}\Pi\Gamma(\tilde{X})$ are the same as the eigenvalues of $\mathcal{D}\Gamma(X)$, except for the $\frac{1}{2}p(p-1)$ eigenvalues equal to one, corresponding with XA for antisymmetric A , which are replaced by zero eigenvalues. The Ostrowski and Ortega-Rheinboldt results now apply directly to $\Pi\Gamma$. If there are no zero distances, if the singular values of X are different, and if the largest eigenvalue of $\mathcal{D}\Pi\Gamma(X)$ is less than one, then X is a point of attraction, the SMACOF iterations converge to X if started sufficiently close to it (in the *attraction ball* of X), and $R_1(X)$ is equal to the largest eigenvalue.

6.2 Rotation to Lower Triangular Form

Instead of rotating to principal components we can also rotate to lower triangular form, in which all $\frac{1}{2}p(p-1)$ elements above the main diagonal of X are zero.

De Leeuw (2023a)

Suppose X_1 , the submatrix with the first p rows of X , is non-singular. We use the QR decomposition of the transpose of X_1 . Thus $X_1' = QR$ with Q square orthonormal and R upper triangular. We have $X_1 = R'Q'$ and thus $X_1Q = R'$. Therefor XQ is lower triangular.

7 Linear Constraints

Instead of $\sum \alpha_s Y_s$ use $x = Y\alpha$.

$$\sigma(\alpha) = 1 - \sum_{1 \leq i < j \leq n} \sum w_{ij} \delta_{ij} d_{ij}(\alpha) + \frac{1}{2} \alpha' Y' V Y \alpha$$

$$\mathcal{D}_s d_{ij}(\alpha) = \frac{1}{d_{ij}(\alpha)} \{Y' A_{ij} Y \alpha\}_s$$

$$\mathcal{D}\sigma(\alpha) = -Y' B(\alpha) Y \alpha + Y' V Y \alpha$$

$$\mathcal{D}^2 \sigma(\alpha) = Y' V Y - (Y' B(\alpha) Y - H(\alpha))$$

$$\Gamma(\alpha) = (Y' V Y)^{-1} Y' B(\alpha) Y \alpha$$

$$\mathcal{D}_s \Gamma(\alpha) = (Y' V Y)^{-1} \{Y' \mathcal{D}_s B(\alpha) Y \alpha + \{Y' B(\alpha) Y\}_s\}$$

$$B(\alpha) = \sum_{1 \leq i < j \leq n} \sum w_{ij} \frac{\delta_{ij}}{d_{ij}(\alpha)} A_{ij}$$

$$\mathcal{D}_s B(\alpha) = - \sum_{1 \leq i < j \leq n} \sum w_{ij} \frac{\delta_{ij}}{d_{ij}^2(\alpha)} A_{ij} \mathcal{D}_s d_{ij}(\alpha)$$

$$\mathcal{D}_s d_{ij}(\alpha) = \frac{1}{d_{ij}(\alpha)} \{Y' A_{ij} Y \alpha\}_s$$

$$\mathcal{D}_s B(\alpha) = - \sum_{1 \leq i < j \leq n} \sum w_{ij} \frac{\delta_{ij}}{d_{ij}^3(\alpha)} A_{ij} \{Y' A_{ij} Y \alpha\}_s$$

$$H(\alpha) := Y' \mathcal{D}_s B(\alpha) Y \alpha = - \sum_{1 \leq i < j \leq n} \sum w_{ij} \frac{\delta_{ij}}{d_{ij}^3(\alpha)} Y' A_{ij} Y \alpha \{Y' A_{ij} Y \alpha\}_s$$

$$\mathcal{D}\Gamma(\alpha) = (Y' V Y)^{-1} \sum_{1 \leq i < j \leq n} \sum w_{ij} \frac{\delta_{ij}}{d_{ij}(\alpha)} \left\{ Y' A_{ij} Y - \frac{Y' A_{ij} Y \alpha \times \alpha' Y' A_{ij} Y}{d_{ij}^2(\alpha)} \right\}$$

Constrained SMACOF

$$X^{(k+1)} = \tilde{V}^+_{\text{tr}} Y'_t V \Gamma(X^{(k)})$$

7.1 Coefficient Space

Also define

$$\{H_{ij}\}_{st} := \text{tr } Y'_s A_{ij} Y_t$$

It is of some interest to look at the special case in which $Y_s = e_k e'_p$ and $Y_t = e_l e'_q$. Then

$$\{H_{ij}\}_{st} = \text{tr } e_p e'_k (e_i - e_j)(e_i - e_j)' e_l e'_q = \delta^{pq}(\delta^{ik} - \delta^{jk})(\delta^{il} - \delta^{jl})$$

It is of some interest to look at the special case in which $Y_s = e_k e'_p$ and $Y_t = e_l e'_q$. Then

$$\{H_{ij}\}_{st} = \text{tr } e_p e'_k (e_i - e_j)(e_i - e_j)' e_l e'_q = \delta^{pq}(\delta^{ik} - \delta^{jk})(\delta^{il} - \delta^{jl})$$

and thus

$$v_{st} = \delta^{pq} \sum_{1 \leq i < j \leq n} \sum w_{ij} (\delta^{ik} - \delta^{jk})(\delta^{il} - \delta^{jl})$$

For the notation in coefficient space we need the square, symmetric, positive semi-definite matrices A_{ij} of order m with elements

$$\{A_{ij}\}_{st} := \text{tr } Y'_s A_{ij} Y_t. \quad (35)$$

The squared distance equals

$$d_{ij}^2(\alpha) := \alpha' A_{ij} \alpha. \quad (36)$$

Note that we continue to overload symbols such as A_{ij} and d . The same is true for V from (5), which we define in coefficient space by using definition (35), and for $\tilde{B}(\alpha)$ from (6). In coefficient space

$$\sigma(\alpha) := 1 - \alpha' B(\alpha) \alpha + \frac{1}{2} \alpha' V \alpha. \quad (37)$$

and the MDS problem is to minimize (37) over $\alpha \in \mathbb{R}^m$.

In coefficient space the Guttman Transform is

$$\Gamma(\alpha) := V^+ B(\alpha) \alpha \quad (38)$$

The *SMACOF algorithm* in configuration space is

$$\alpha^{(k)} = \Gamma(\alpha^{(k-1)}) = \Gamma^k(\alpha^0). \quad (39)$$

8 Accelerated SMACOF

Suppose $\text{Gamma}(X) = X$ and $\tau \neq 1$. Then $2\Gamma(\tau X) - \tau X = (2 - \tau)X$ and $2\Gamma((2 - \tau)X) - (2 - \tau)X = \tau X$ Also $\sigma(\tau X) = \sigma((2 - \tau)X)$

9 Software

The appendix has an ad-hoc version of the `smacof` function. Its arguments are

```
args(smacof)
```

```
function (delta, w, p = 2, xold = torgerson(delta, p), pca = FALSE,  
         verbose = FALSE, eps = 1e-15, itmax = 10000)  
NULL
```

The function expects both dissimilarities and weights to be of class `dist`. The default initial configuration uses classical MDS (Torgerson (1958)). We iterate until either the distance $\eta(X^{(k)} - X^{(k-1)})$ between successive configurations is less than `eps`, which has the ridiculously small default value of 10^{-15} , or until the maximum number of iterations `itmax` is reached, which defaults to the ridiculously large default value of 10000. If `pca` is `TRUE` all iterates are rotated to principal components. If `verbose` is `TRUE` we print, for each iteration, the iteration number k , the stress $\sigma(X^{(k)})$, the change $\eta(X^{(k)} - X^{(k-1)})$, and the root and ratio convergence factors.

The second file `derivative.R` computes the Jacobian of the iteration functions at the limit (i.e. at the point where SMACOF stops). There are actually six functions in the file. The three functions `dGammaA`, `dPiA` and `dPiGammaA` use the analytical expressions we have derived earlier for $\mathcal{D}\Gamma(X)$, $\mathcal{D}\Pi(X)$ and $\mathcal{D}\Pi\Gamma(X)$. The corresponding functions `dGammaN`, `dPiN` and `dPiGammaN` numerically compute the Jacobian using the `jacobian` function from the `numDeriv` package (Gilbert and Varadhan (2019)). They are basically used to check the formulas.

10 Examples

10.1 Ekman

Time for a numerical example. We will use the color similarity data of Ekman (1954), taken from the SMACOF package (De Leeuw and Mair (2009)). We transform Ekman's similarities s_{ij} to dissimilarities δ_{ij} using $\delta_{ij} = (1 - s_{ij})^3$, and we use unit weights in W .

SMACOF requires 52 iterations for convergence. The minimum of stress is 0.0110248119, the root factor is equal to 0.5085800483 and the ratio factor is 0.5329246268. The eigenvalues of $\mathcal{D}\Gamma(X)$ are

```

0.9999999999999999
0.538510668196408
0.532498554224166
0.529669334191043
0.525541097754551
0.519602718218807
0.516533687841054
0.513727908691608
0.510295310409166
0.506357695934493
0.495649713446001
0.481518780073304
0.469548625536451
0.445038589020916
0.410479252654484
0.394284315478173
0.357670750114585
0.348395413045201
0.330341477528788
0.313520384427863
0.287598015841956
0.280399104121623
0.267278474596759
0.247631495462991
0.216576009083047
0.0000000000000000
0.0000000000000000
0.0000000000000000

```

As an aside, the Ekman example (with this particular transformation of the similarities) is special, because the two-dimensional solution is actually the global minimum over all configurations (i.e. the global minimum of MDS for all $1 \leq p \leq n$). As shown in De Leeuw (2014) this follows from the fact that the two unit eigenvalues of $V^+B(X)$ are actually its two largest eigenvalues, and consequently X is the solution of the convex full-dimensional nMDS relaxation of the MDS problem (De Leeuw et al. (2016)). The eigenvalues of $V^+B(X)$ are

```

1.0000000000000000
1.0000000000000000
0.923497086367287
0.907901212970889

```

0.862936584878895
0.852692003103344
0.829803620857357
0.814556167662381
0.793238576338502
0.791651722456649
0.786442678118770
0.747679475705025
0.728268247434340
0.000000000000001

We now set `pca=TRUE` and run `mSMACOF`. It requires 51 iterations for convergence. The minimum of stress is 0.0110248119, and the root factor is equal to 0.5080118475 and the ratio factor is 0.5718672565. The eigenvalues of $\mathcal{D}\Pi\Gamma(X)$ at the fixed point X are

0.538510668196407
0.532498554224166
0.529669334191045
0.525541097754552
0.519602718218807
0.516533687841055
0.513727908691609
0.510295310409168
0.506357695934493
0.495649713446002
0.481518780073306
0.469548625536451
0.445038589020917
0.410479252654486
0.394284315478174
0.357670750114585
0.348395413045201
0.330341477528788
0.313520384427864
0.287598015841957
0.280399104121623
0.267278474596758
0.247631495462991
0.216576009083047
0.000000000000000

0.0000000000000000
0.0000000000000000
0.0000000000000000

They are equal to the eigenvalues of $\mathcal{D}\Gamma(X)$ and the root convergence factor $R_1(\{X^{(k)}\})$ is 0.5385106682.

10.2 De Gruijter

The second example are dissimilarities between nine Dutch political parties, collected a long time ago by De Gruijter ([1967](#)).

	KVP	PvdA	VVD	ARP	CHU	CPN	PSP	BP
PvdA	2.63							
VVD	2.27	3.72						
ARP	1.60	2.64	2.46					
CHU	1.80	3.22	1.97	0.20				
CPN	4.54	2.12	5.13	4.84	4.80			
PSP	3.73	1.59	4.55	3.73	4.08	1.08		
BP	4.18	4.22	3.90	4.28	3.96	3.34	3.88	
D66	3.17	2.47	1.67	3.13	3.04	4.42	3.36	4.36

We analyze the data with SMACOF using three dimensions.

SMACOF requires 779 iterations for convergence. The minimum of stress is 0.003442194, the root factor is equal to 0.9565946372, and the ratio factor is 0.9684304863. The eigenvalues of $\mathcal{D}\Gamma(X)$ are

1.0000000000000001
1.0000000000000000
1.0000000000000000
0.965505429805660
0.940592046981168
0.919047686446446
0.863993920924431
0.822263696226349
0.810020971414307
0.771947328045071
0.728539213663602

0.712621684571535
 0.659318624263222
 0.638485365688918
 0.624552464148482
 0.616085432872672
 0.557480497708779
 0.501954186928526
 0.458630667850014
 0.450598367846591
 0.355243400669833
 0.309186479174062
 0.247708397109091
 0.000000000000002
 0.000000000000002
 0.000000000000000
 0.000000000000000

In this case there is no guarantee that we have the three-dimensional global minimum. The unit eigenvalues of the matrix $V^+B(X)$ are not the three largest ones.

1.079524009371955
 1.032606649163671
 1.000000000000000
 1.000000000000000
 1.000000000000000
 0.986706272372898
 0.971839080877692
 0.906211919383164
 0.000000000000001

mSMACOF in three dimensions requires 778 iterations for convergence. The minimum of stress is 0.003442194, the root factor is equal to 0.9565709099, and the ratio factor is 0.9714170285. The eigenvalues of $\mathcal{D}\Pi\Gamma(X)$ are

0.965505429805661
 0.940592046981168
 0.919047686446446
 0.863993920924433
 0.822263696226351

0.810020971414307
0.771947328045072
0.728539213663603
0.712621684571534
0.659318624263222
0.638485365688918
0.624552464148483
0.616085432872671
0.557480497708778
0.501954186928526
0.458630667850015
0.450598367846592
0.355243400669831
0.309186479174061
0.247708397109091
0.0000000000000001
0.0000000000000001
0.0000000000000001
0.0000000000000000
0.0000000000000000
0.0000000000000000
0.0000000000000000
0.0000000000000000

Again, as expected, the eigenvalues for SMACOF and mSMACOF are the same and the root convergence factor is the spectral norm of the mSMACOF derivative, which is 0.9655054298.

11 Conclusion

From a practical point of view there is no need to ever use modified SMACOF. We only use the Guttman transform, and compute the largest eigenvalue of its derivative at the solution X , ignoring the $\frac{1}{2}p(p-1)$ trivial unit eigenvalues. That largest eigenvalue, if it is strictly smaller than one, is the root convergence factor. In that case SMACOF can be said to converge to $\Pi(X)$, which is the SMACOF solution rotated to principal components.

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